

Simulating disease transmission

(Part 3)

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SCIENCE AND
EDUCATION
FOR
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LIFE

Overview

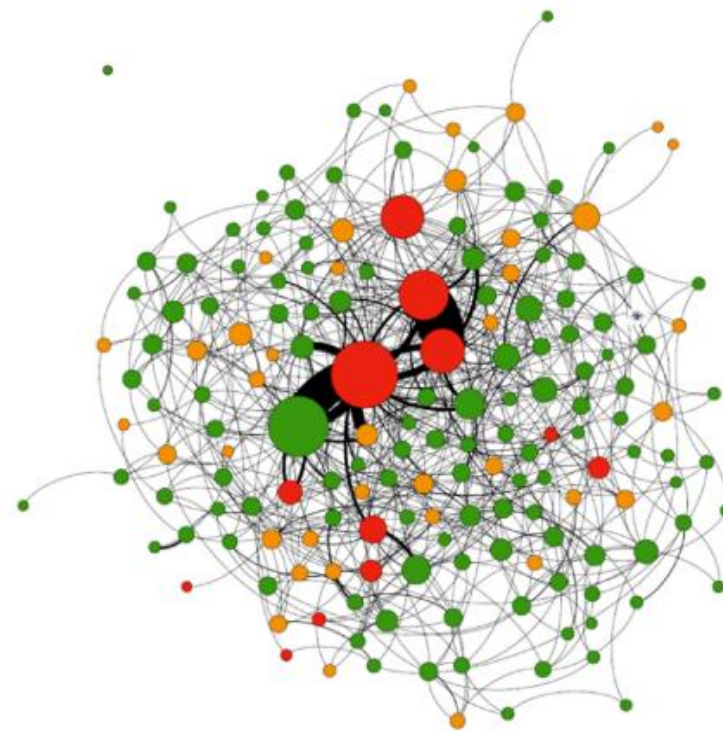
This lecture will be in 3 parts:

1. Network-based transmission models
2. Introduction to SimNet
3. Incorporating breeding values into transmission models

1. Network-based transmission models

1. Network-based transmission models

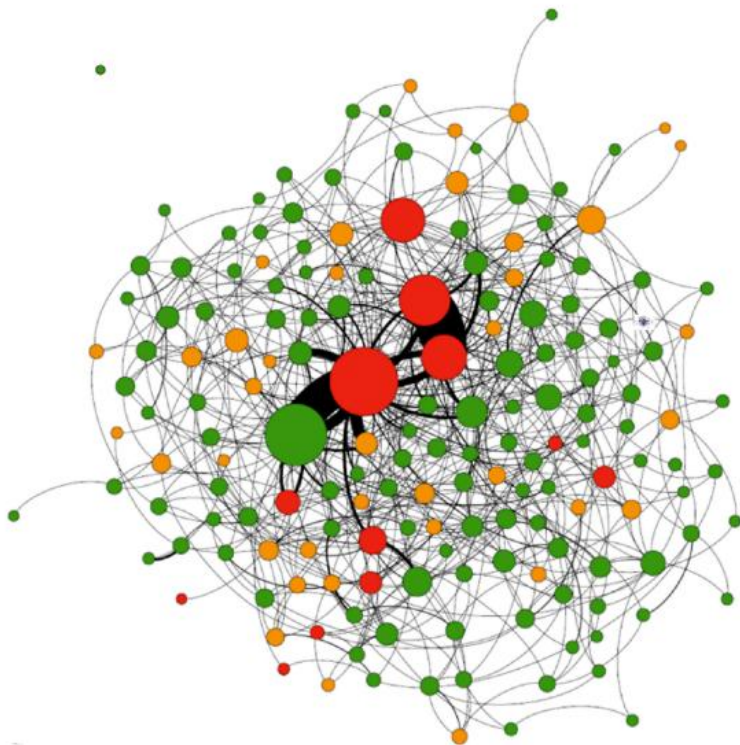
- A **network** can be constructed from **relational data** and can be defined as a set of **entities**, such as people, groups, and organizations, with some relationships or interactions between them.



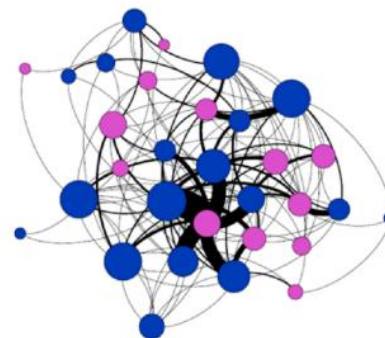
(de Freslon et al., 2019)

1. Network-based transmission models

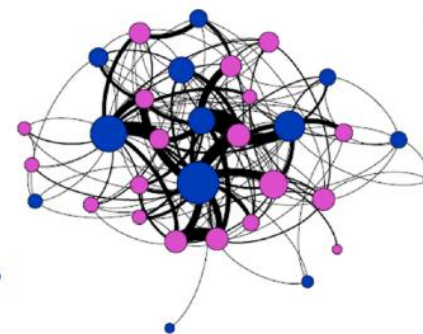
COW



CALF1



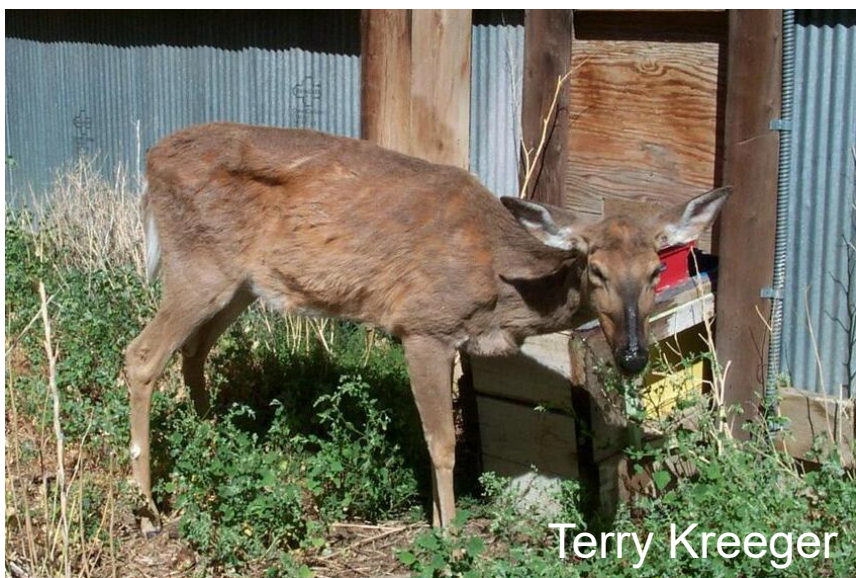
CALF2



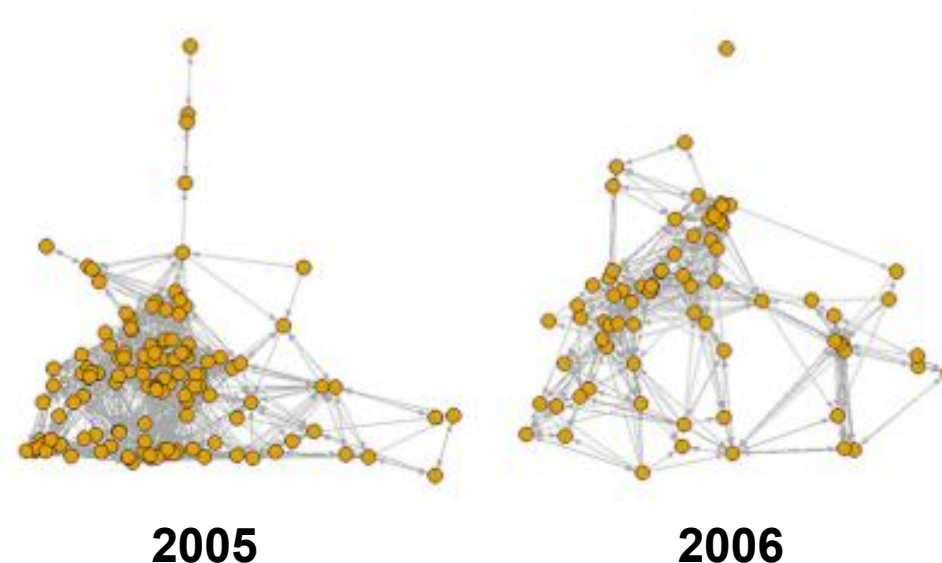
Focused on three contact behaviours that may lead to transmission of pathogenic *Leptospira* spp.: **sniffing, licking and rubbing the face** on the genital area of another animal

(de Freslon et al., 2019)

1. Network-based transmission models



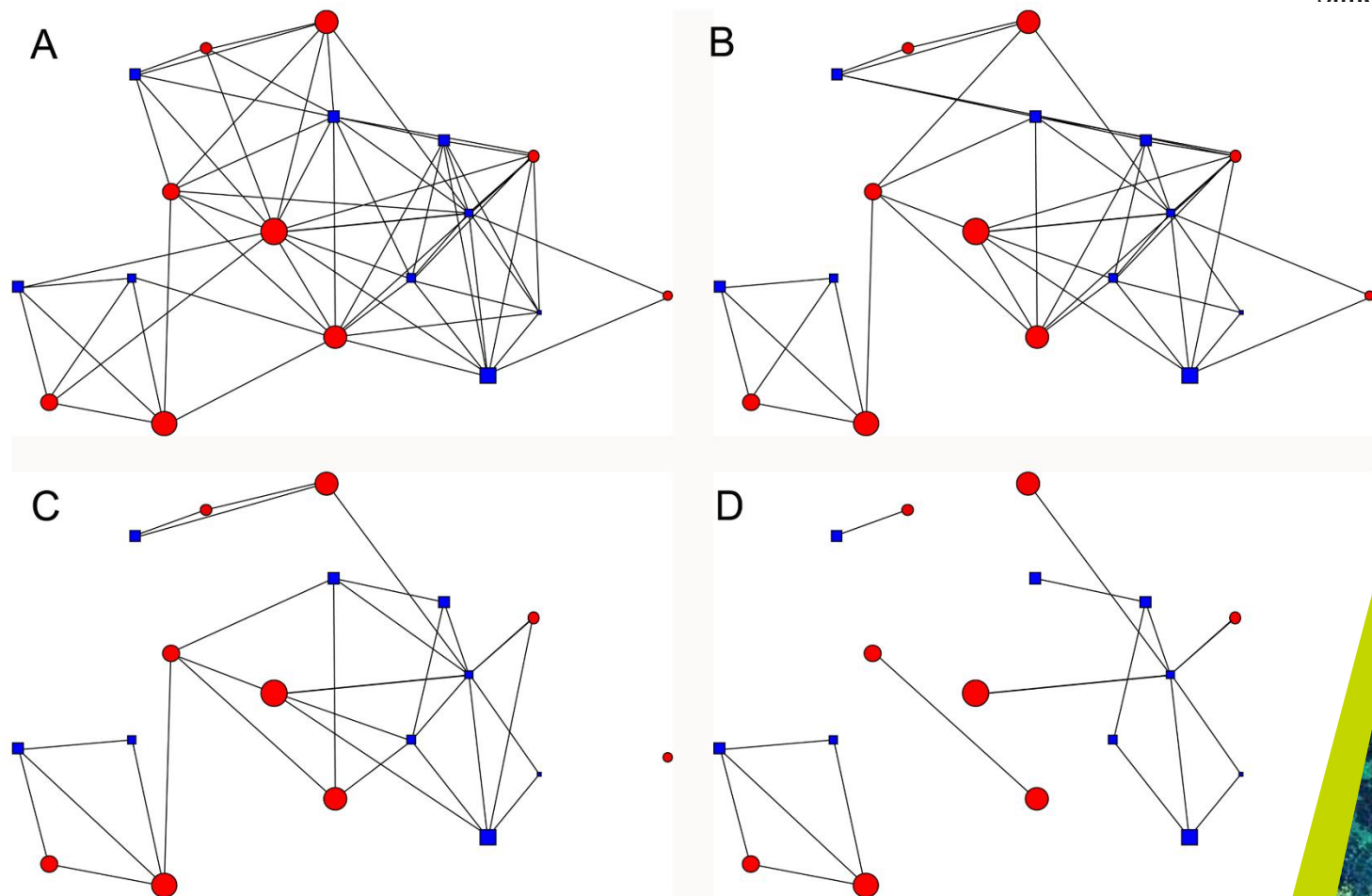
Unweighted



Studies that estimate the role of **scrape-related contact** relative to other types of indirect contact or direct contact will be important for **understanding the potential impacts on chronic wasting disease (CWD)** transmission dynamics.

(Egan et al., 2023)

1. Network-based transmission models



These findings demonstrate that **raccoon populations are much more tightly connected** than would have been predicted based on previous studies, but also point out that additional research is needed to calculate more precise transmission probabilities by infected individuals (Rabies, canine distemper, parvovirus, leptospirosis, etc.). (Hirsch et al., 2013)

1. Network-based transmission models

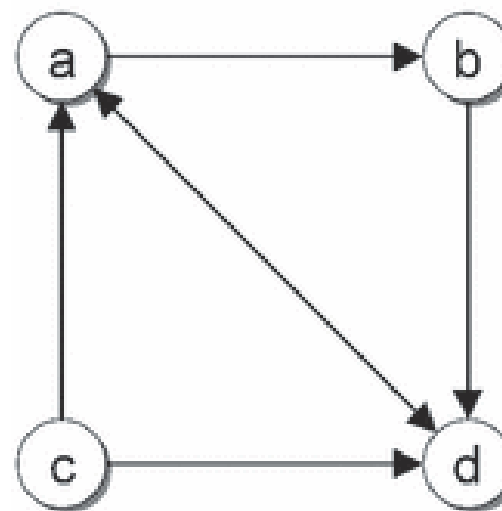
Adjacency matrix

	a	b	c	d
a	0	1	0	1
b	0	0	0	1
c	1	0	0	1
d	1	0	0	0

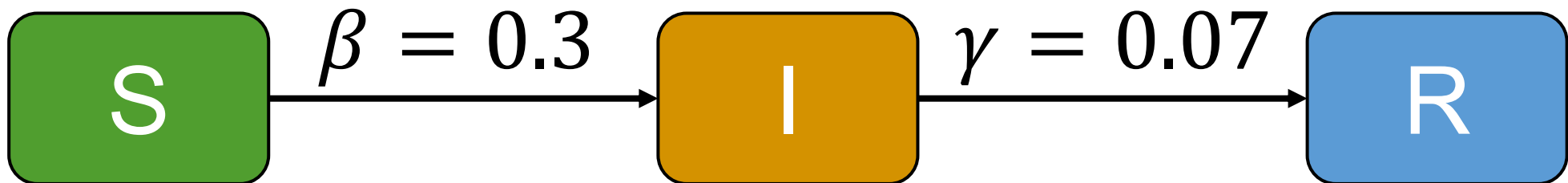
Notation

$$G = \{(a, b), (a, d), (b, d), (c, a), (c, d), (d, a)\}$$

Graph



1. Network-based transmission models



Spatial transmission models

Network transmission models

$$p_{ij} = \beta e^{-d_{ij}/e}$$

Distance

Spread

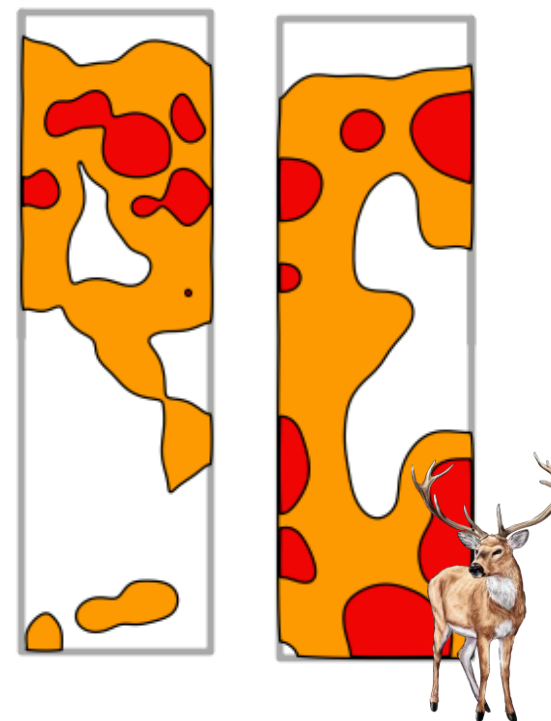
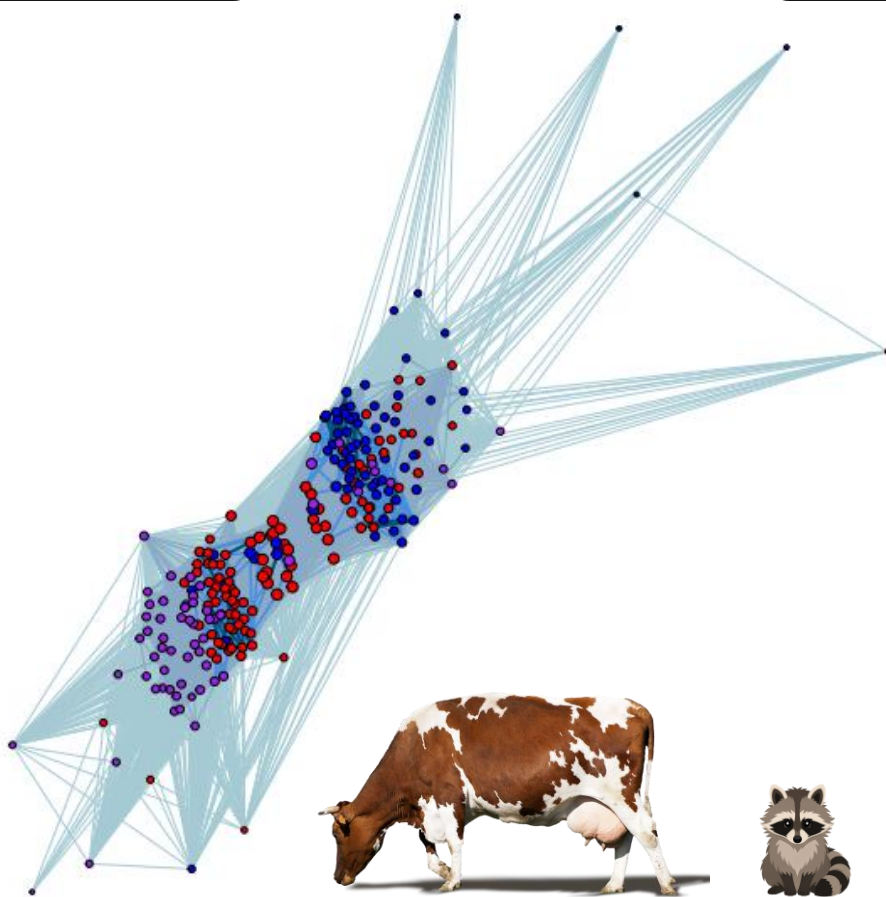
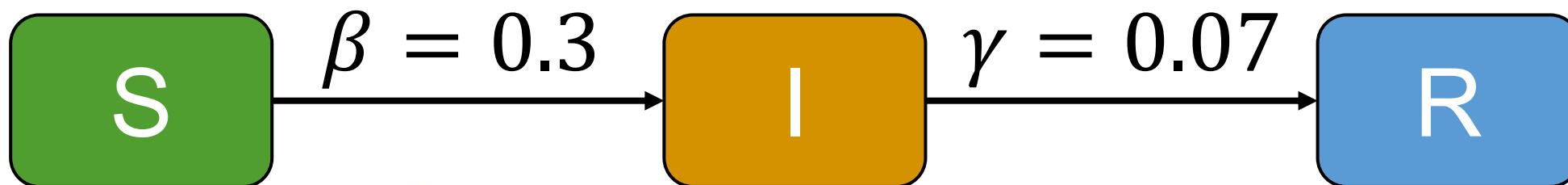
$$P(\text{infection}) = 1 - \prod_{j=1}^k (1 - p_{ij})$$

Network value

$$p_{ij} = 1 - e^{-\beta \phi_{ij,t}}$$

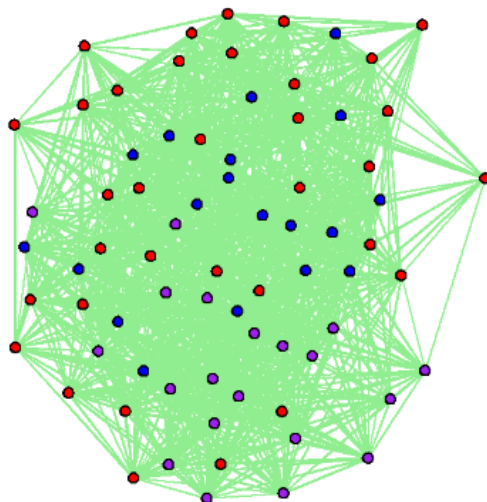
$$P(\text{infection})_{it} = 1 - \prod_{j=1}^k (1 - p_{ij,t})$$

1. Network-based transmission models

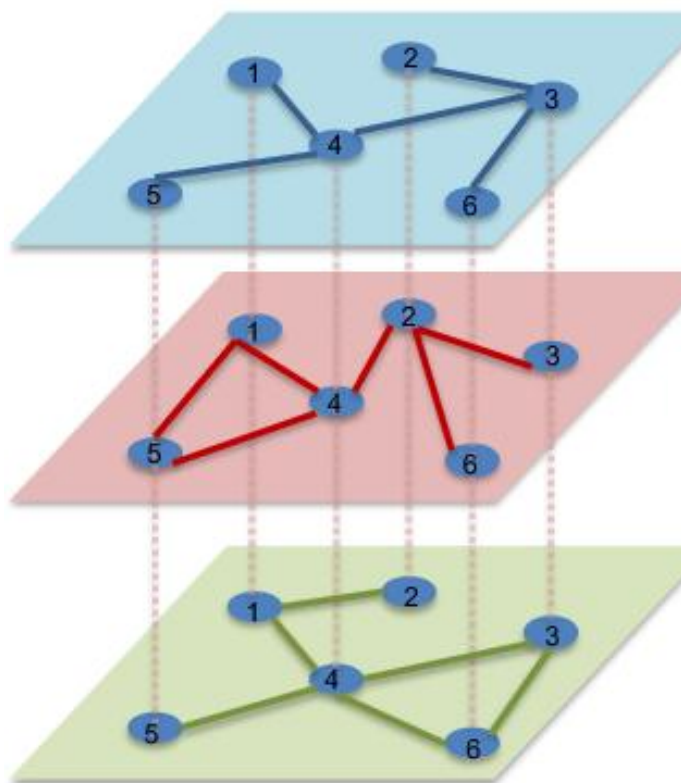


1. Network-based transmission models

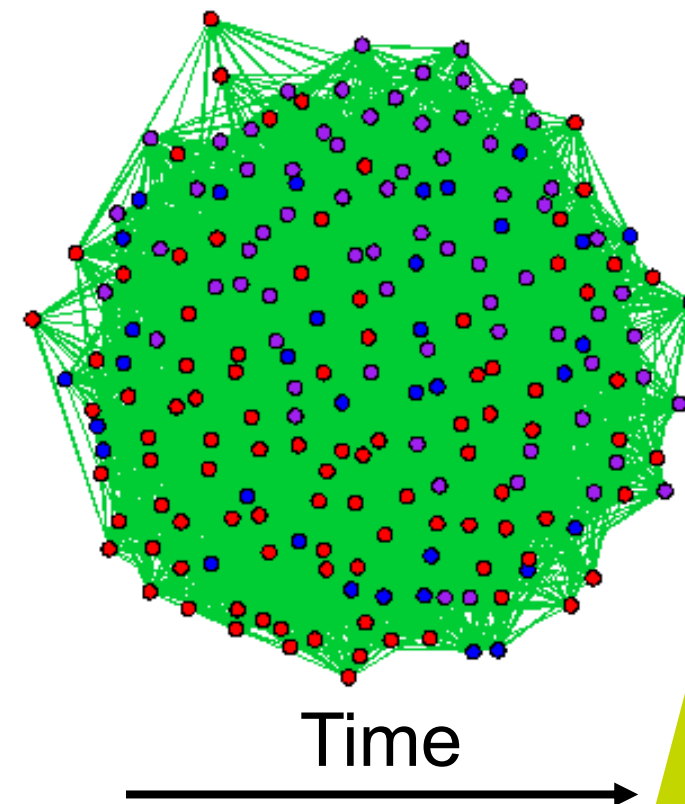
Unidimensional data



Multidimensional data

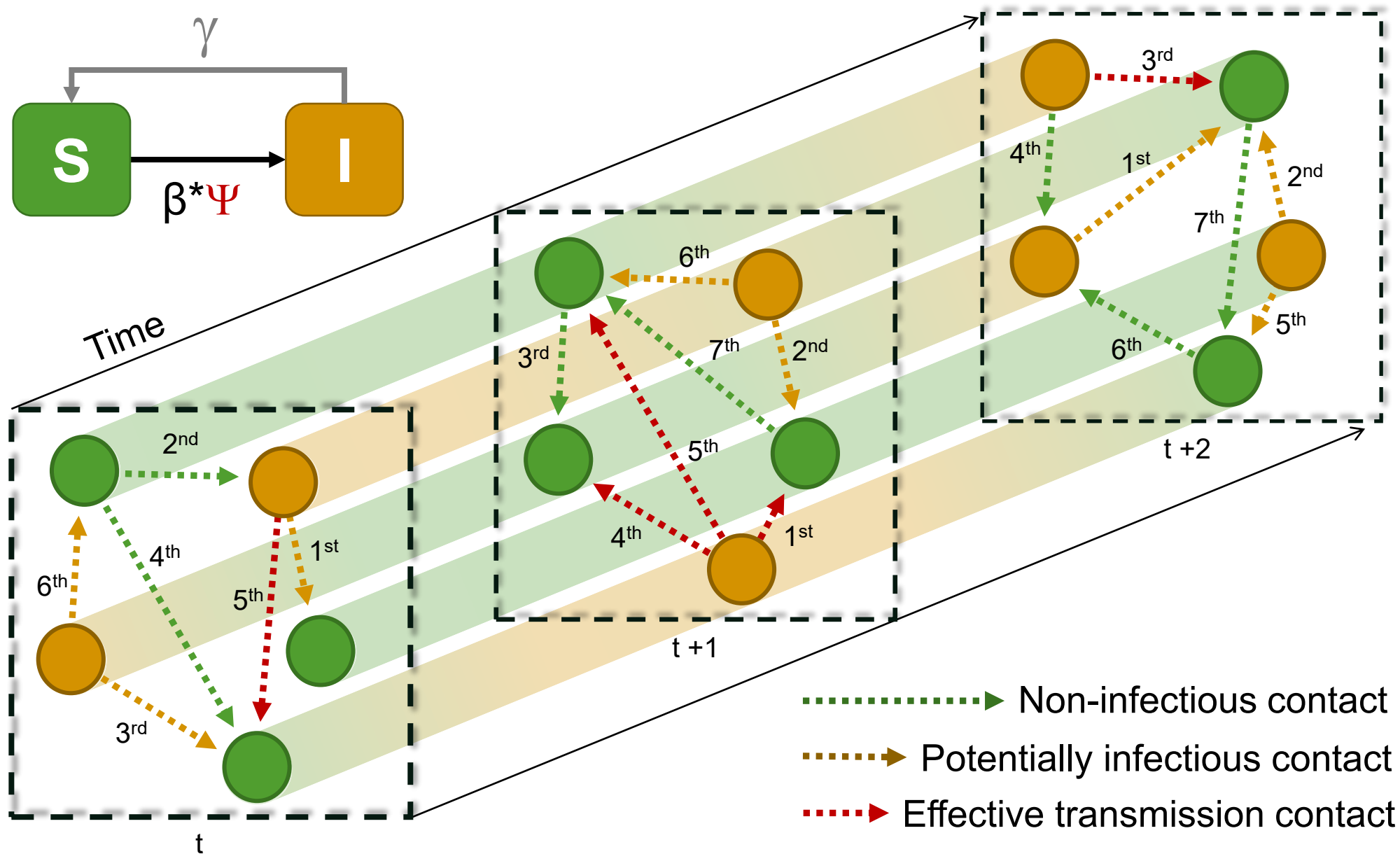


Longitudinal data

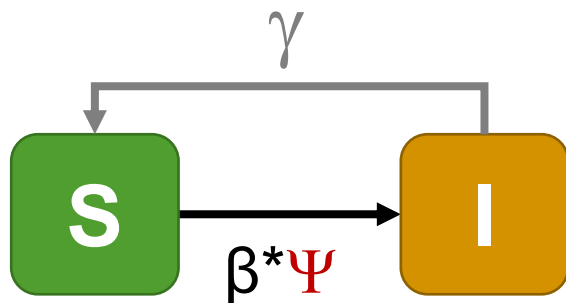


2. Introduction to *SimNet*

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Naïve transmission

β : 0.75

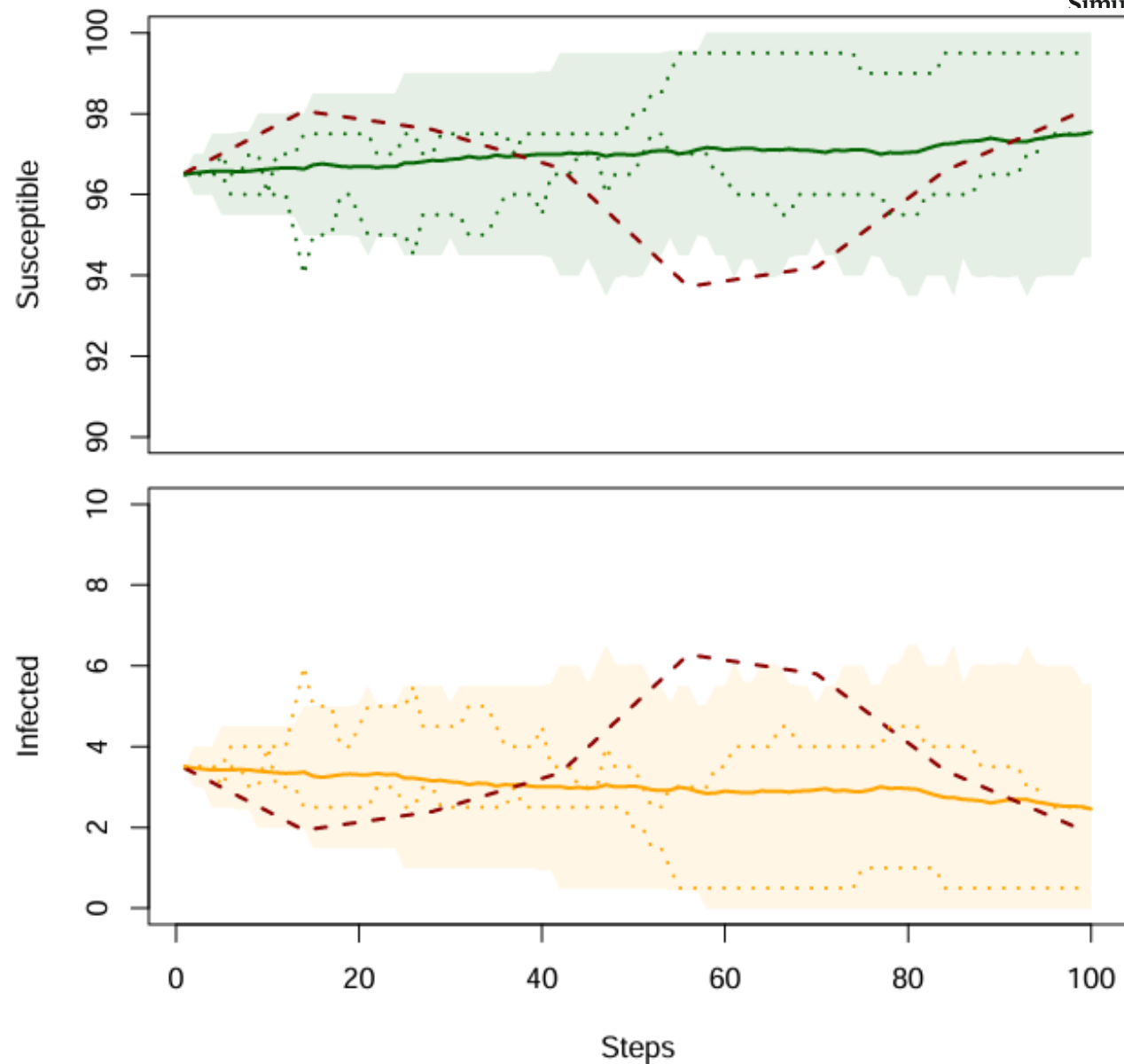
γ : 0.58

Ψ : 1

Ind. Inf₁ = 7

Total Ind. = 200

Replicates = 100



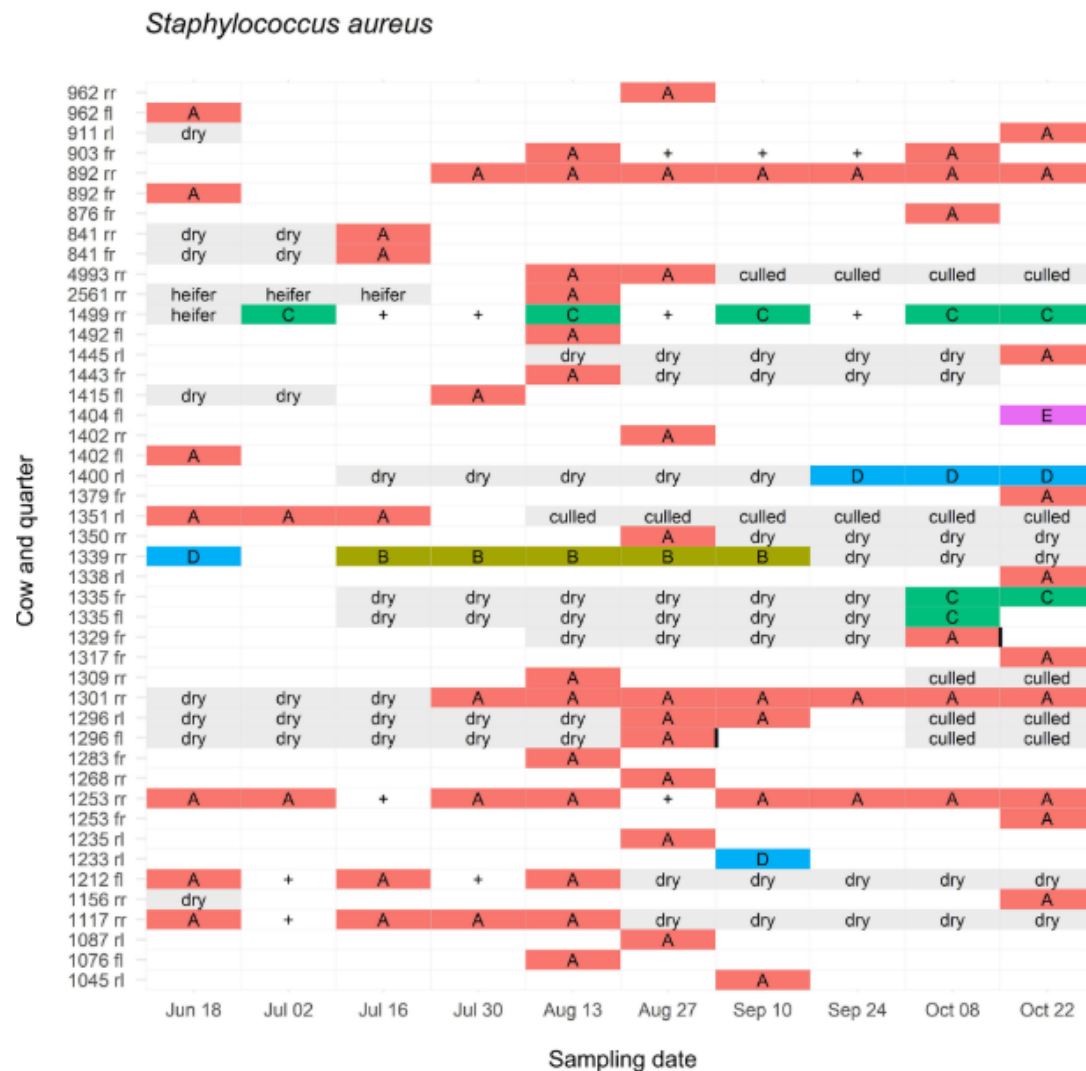
UNIVERSITY OF COPENHAGEN
DEPARTMENT OF VETERINARY AND ANIMAL SCIENCES

Transmission pathways and infection durations of bovine mastitis pathogens

PhD Thesis 2023

Svenja Woudstra

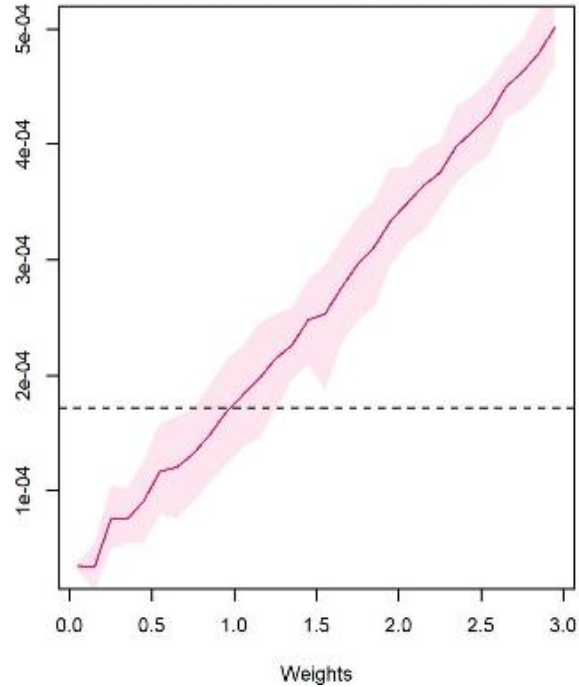
This thesis has been submitted to the Graduate School of Health and Medical Sciences, University of Copenhagen, 31st of May 2023



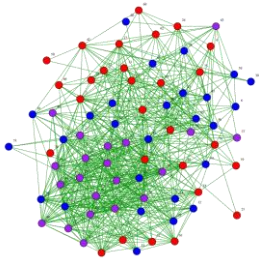
(Woudstra et al., 2017)

2. Introduction to *SimNet*

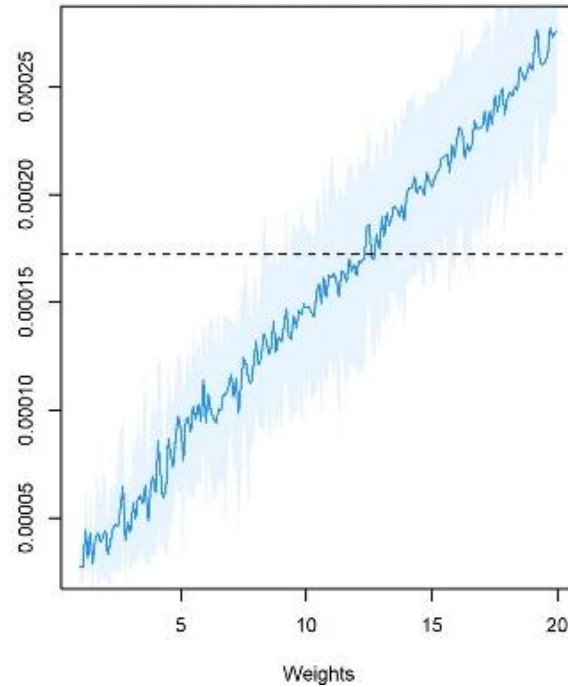
Direct transmission networks (SIN and ASN)



$\Psi : 0.45$



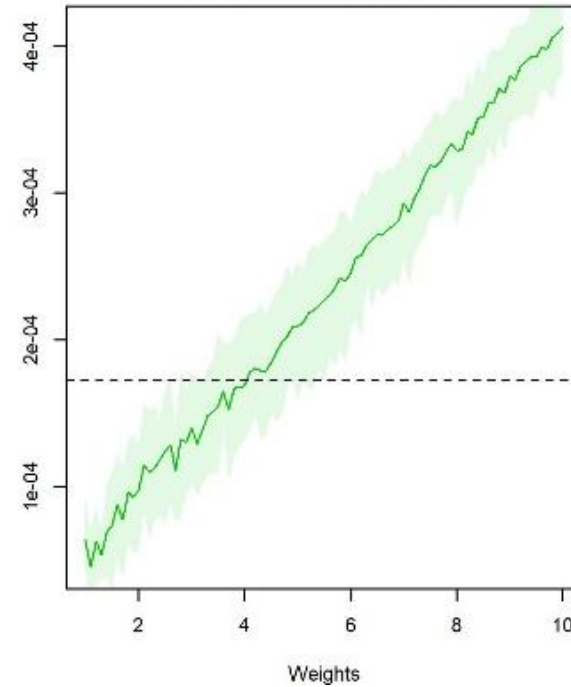
Milking parlour network (MPN)



$\Psi : 12$



Resting cubicle network (RCN)



$\Psi : 4$



2. Introduction to *SimNet*



Applications of *SimNet*

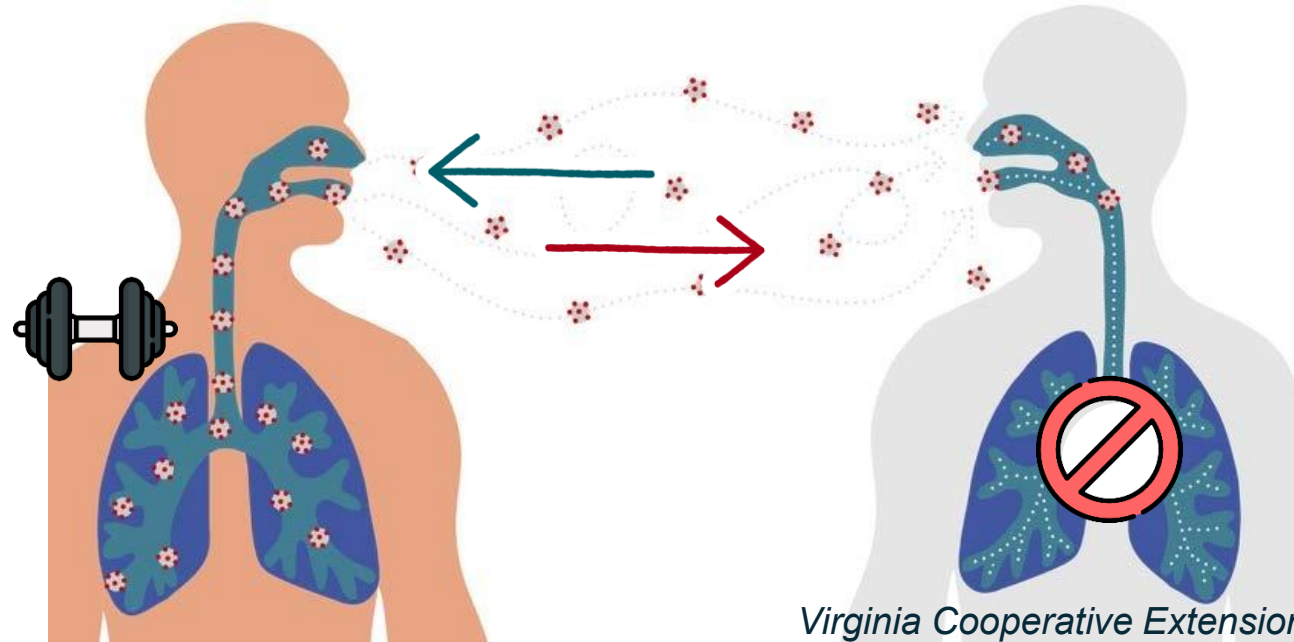
- **Simulate disease transmission** through temporal direct, indirect, or homogeneous contact networks.
- **Evaluate sampling periods and temporal resolution** for studying transmission dynamics.
- **Compare alternative contact networks** and identify the most plausible route of transmission.
- **Identify likely transmission events and infection sources** at the individual level.
- **Estimate transmission parameters** and generate information that can be used to quantify individual **susceptibility and infectivity**.
- **Test alternative epidemiological scenarios**, including changes in population structure, contact patterns, and disease parameters.

3. Incorporating breeding values into transmission models

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← **Susceptibility:** propensity to become infected and/or develop disease.

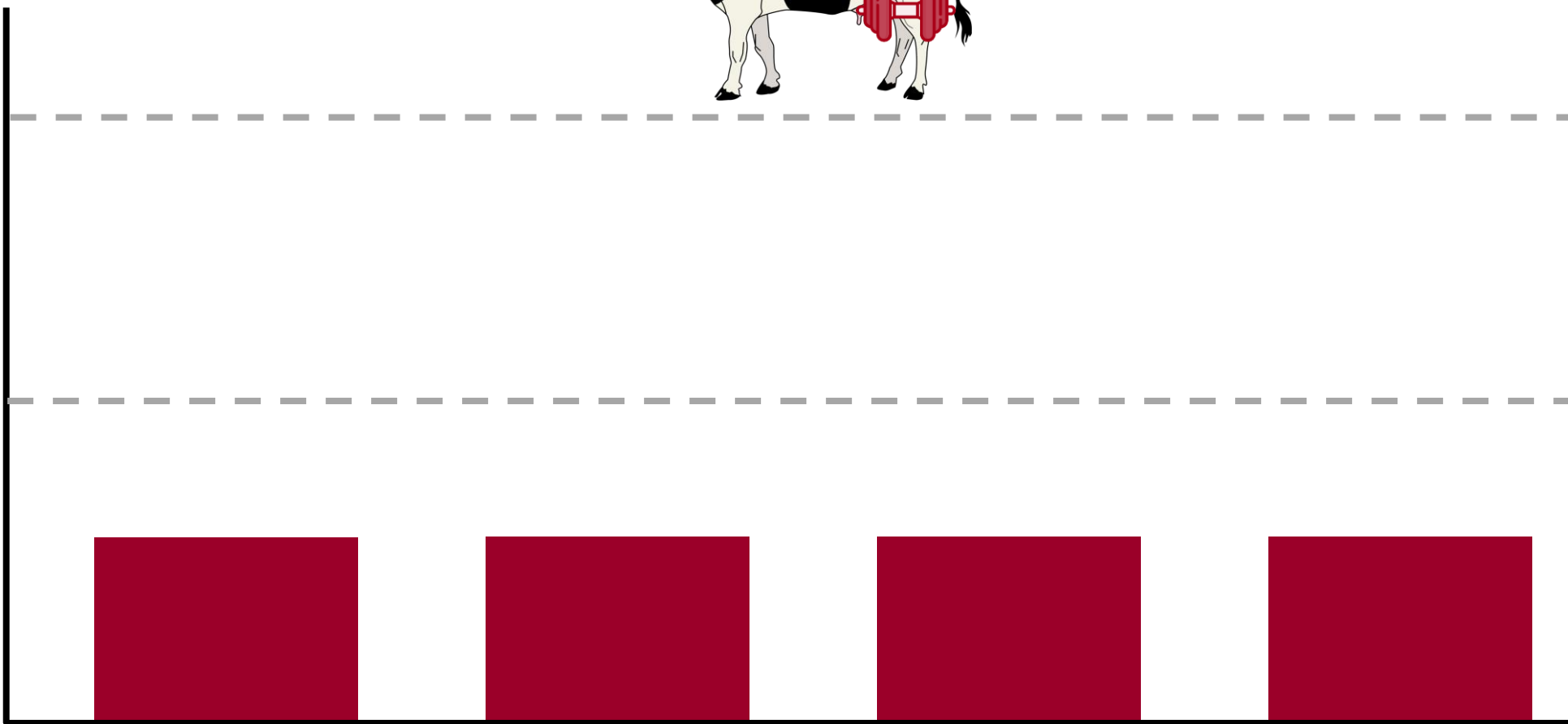
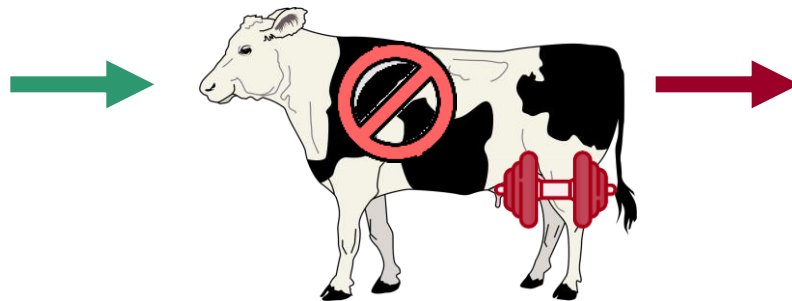
→ **Infectivity:** propensity of an infected individual to transmit infection



 **Resilience:** capacity to be minimally affected by exposure to infection and/or maintain high production performance.

 **Resistance:** Ability of the individual to inhibit or limit within-host pathogen replication

(Doeschl-Wilson et al., 2021)



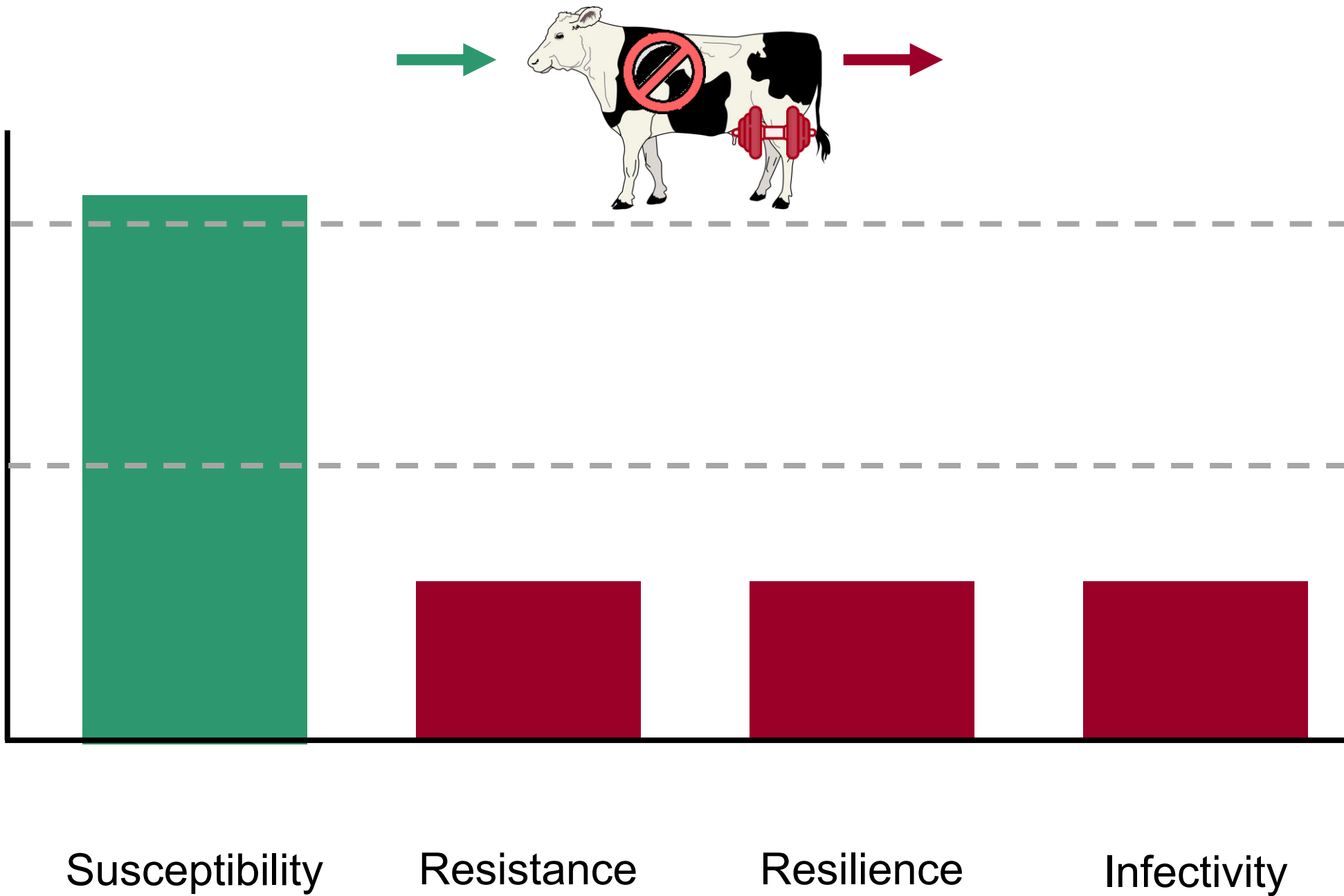
Susceptibility

Resistance

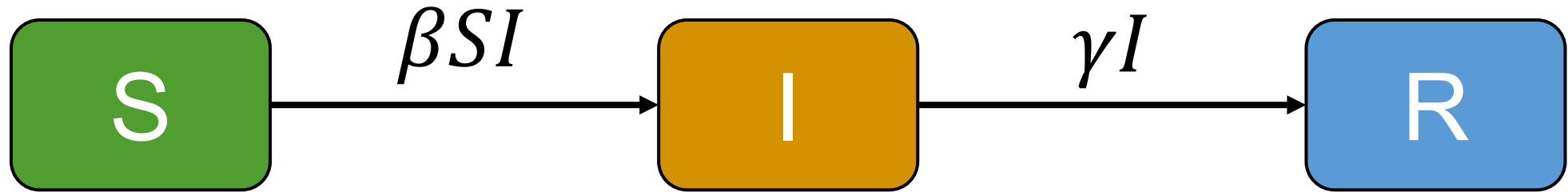
Resilience

Infectivity



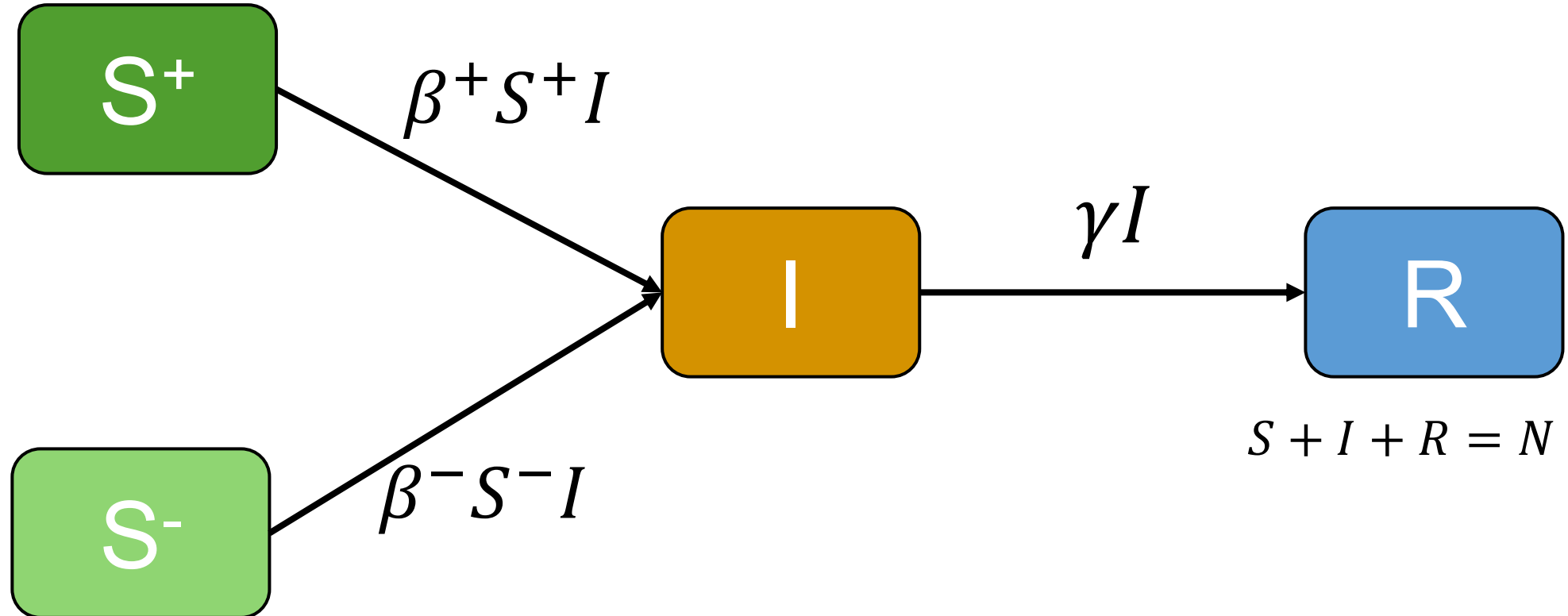


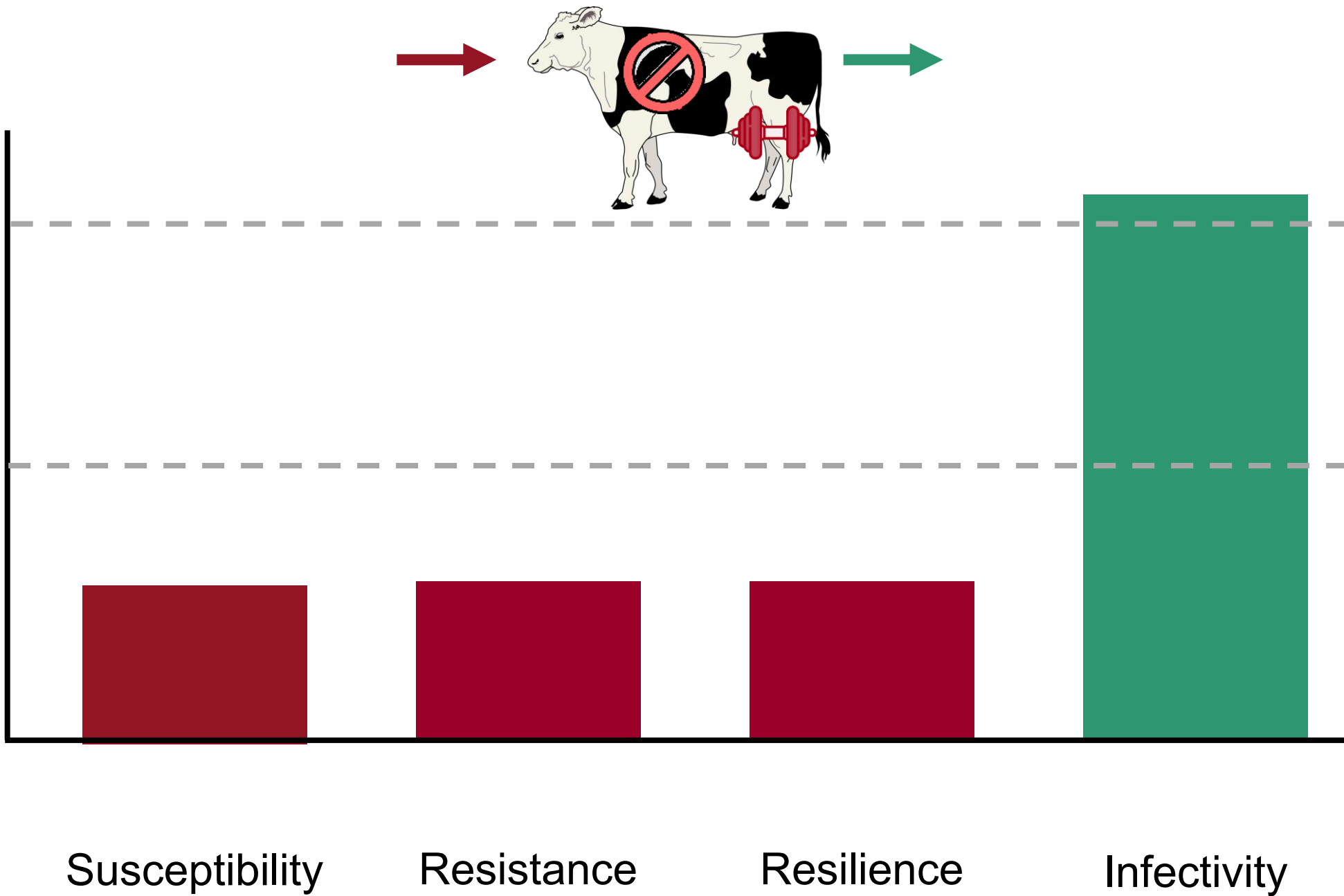
3. Incorporating breeding values into transmission models



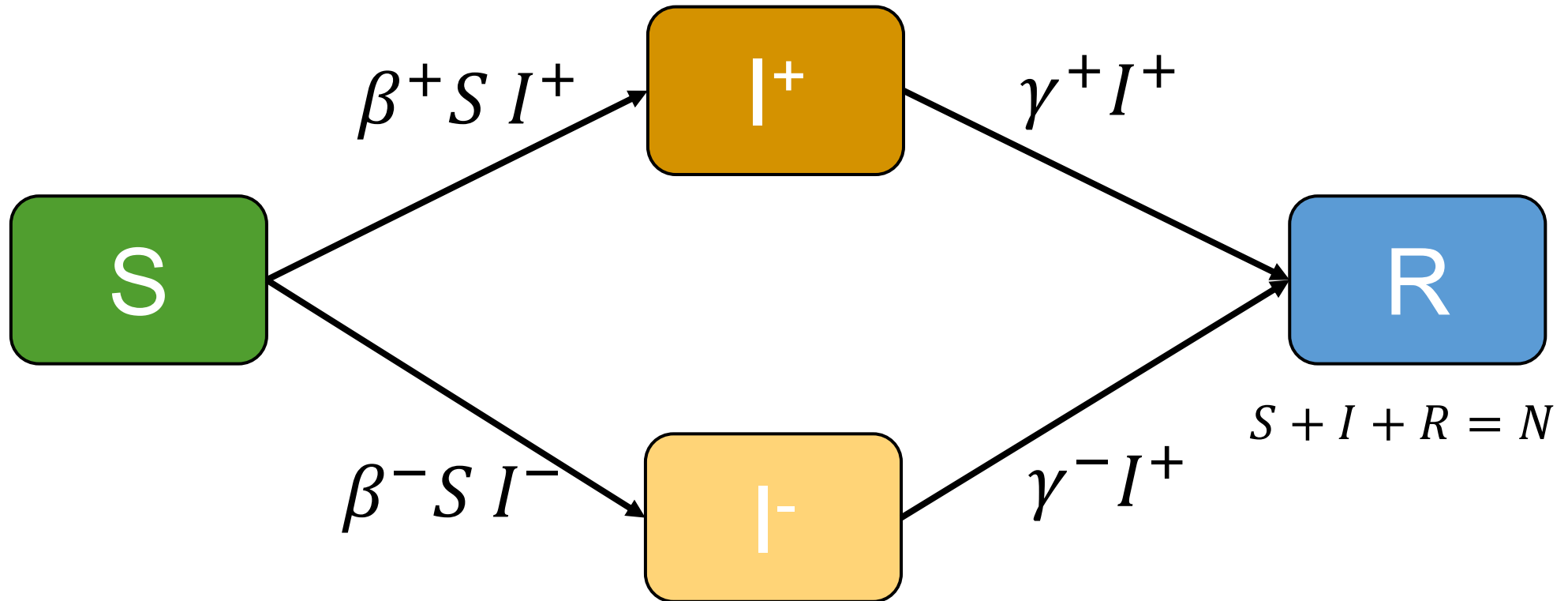
$$S + I + R = N$$

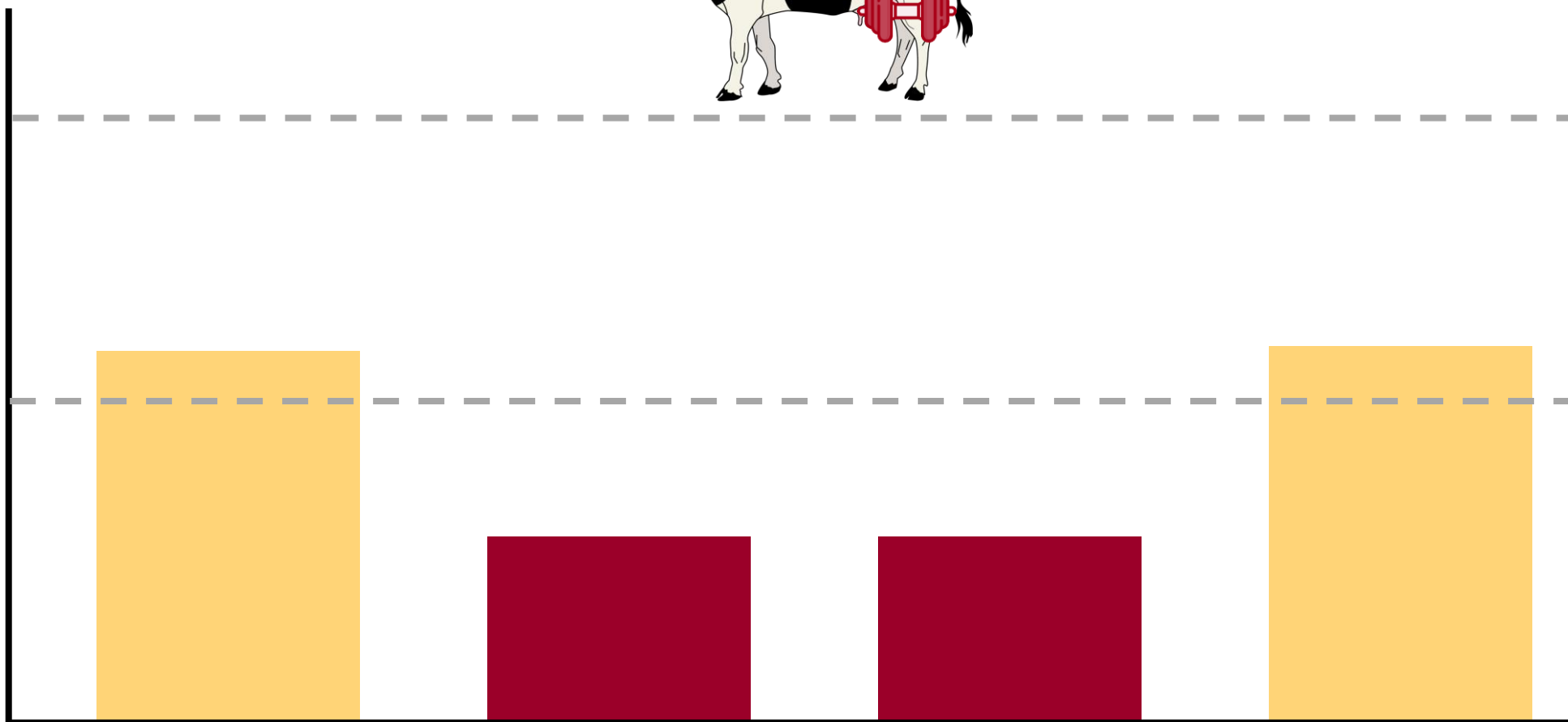
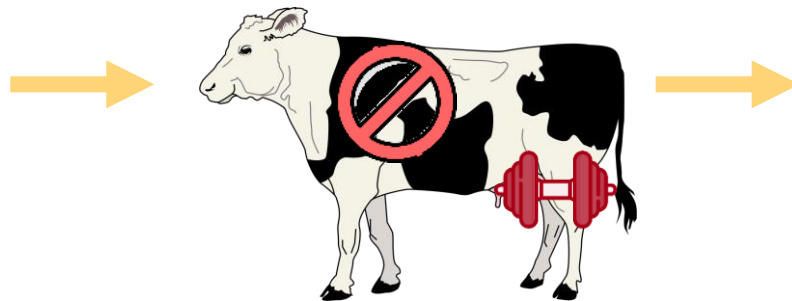
3. Incorporating breeding values into transmission models





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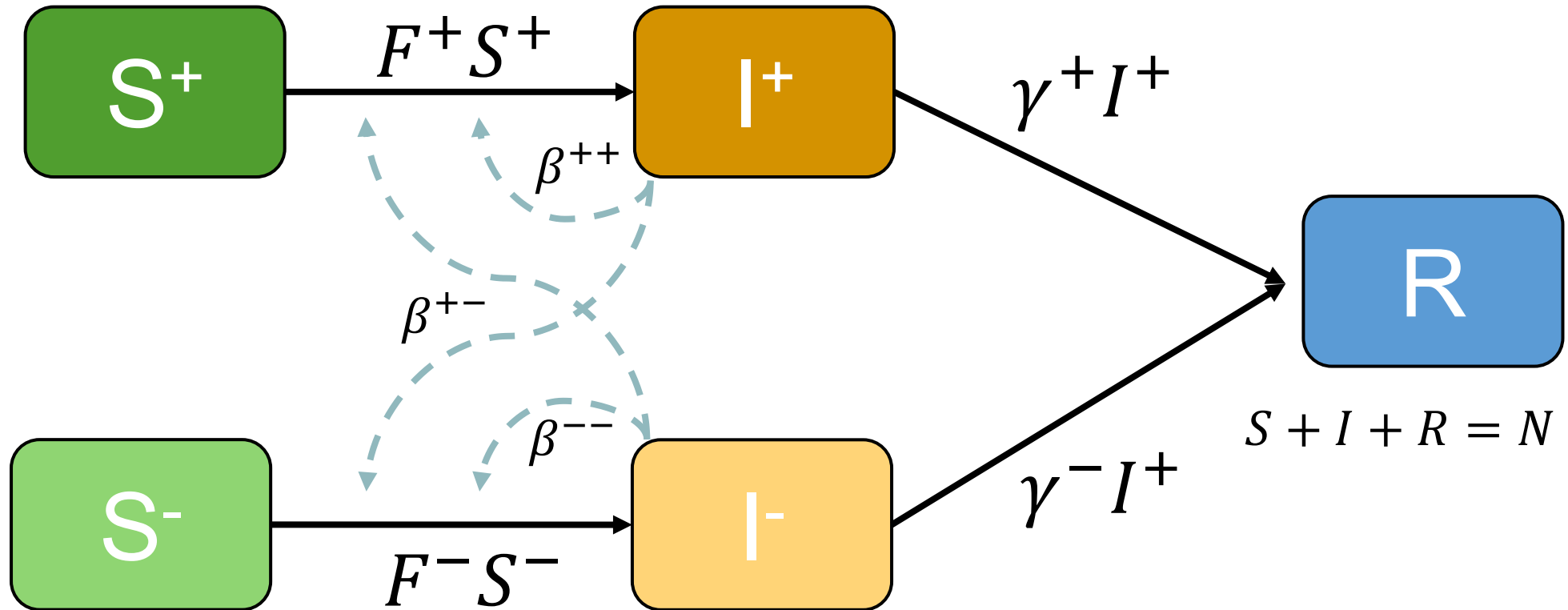
Susceptibility

Resistance

Resilience

Infectivity

3. Incorporating breeding values into transmission models



Force of infection:

$$F^+ = (\beta^{++}I^+ + \beta^{+-}I^-)/N$$

$$F^- = (\beta^{+-}I^+ + \beta^{--}I^-)/N$$

Who infects who?

$$\boldsymbol{\beta} = \begin{pmatrix} \beta^{++} & \beta^{+-} \\ \beta^{+-} & \beta^{--} \end{pmatrix}$$

3. Incorporating breeding values into transmission models

- Modelling framework – Formulation

- **Deterministic models:**



Predicts a single outcome for a given set of parameters and initial conditions.

- **Stochastic models:**



Includes chance, so the same parameters can produce different outcomes.

- Modelling framework – Population representation

- **Compartmental models :**

Groups individuals into population states, such as susceptible, infected, or recovered.

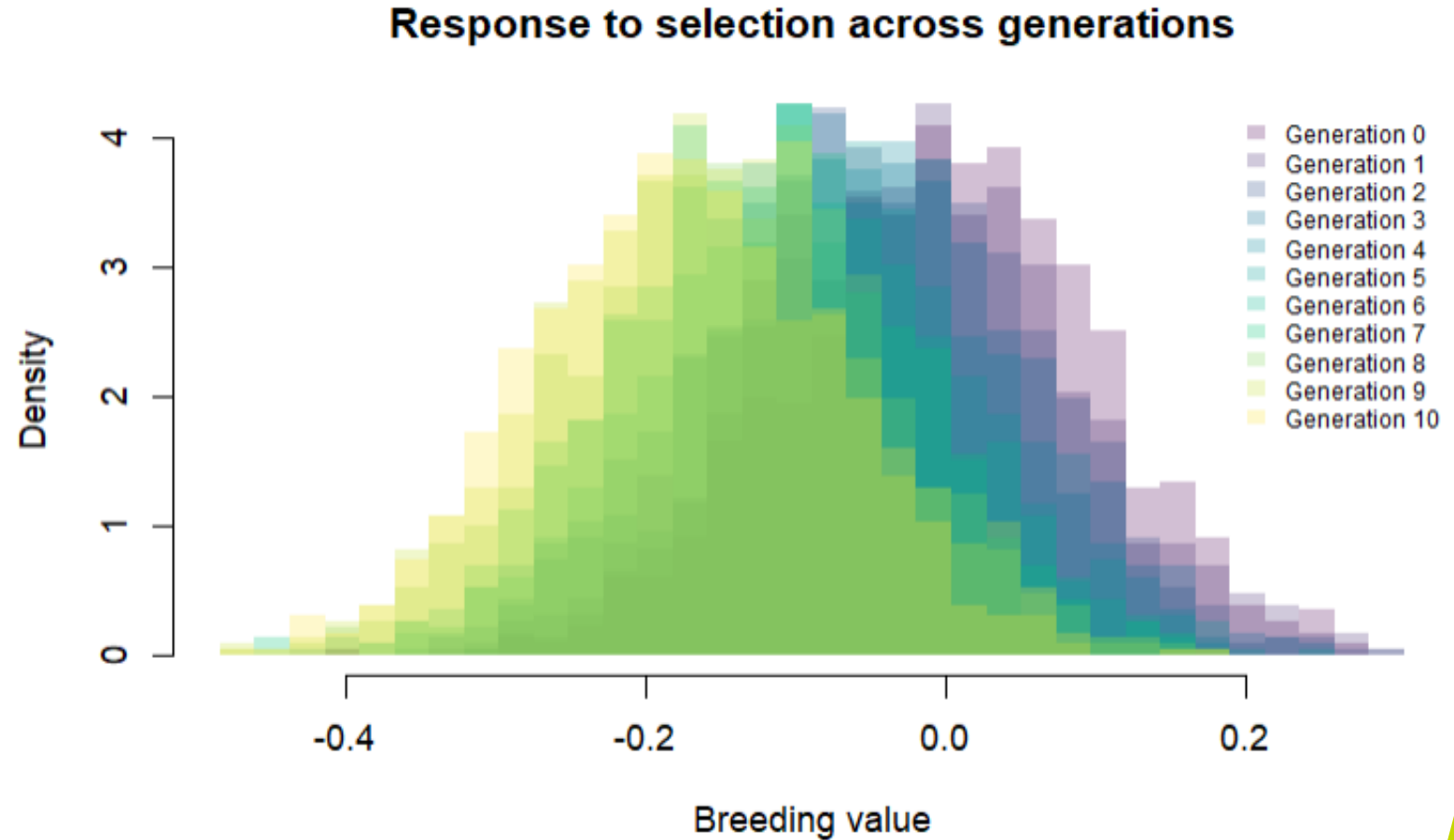
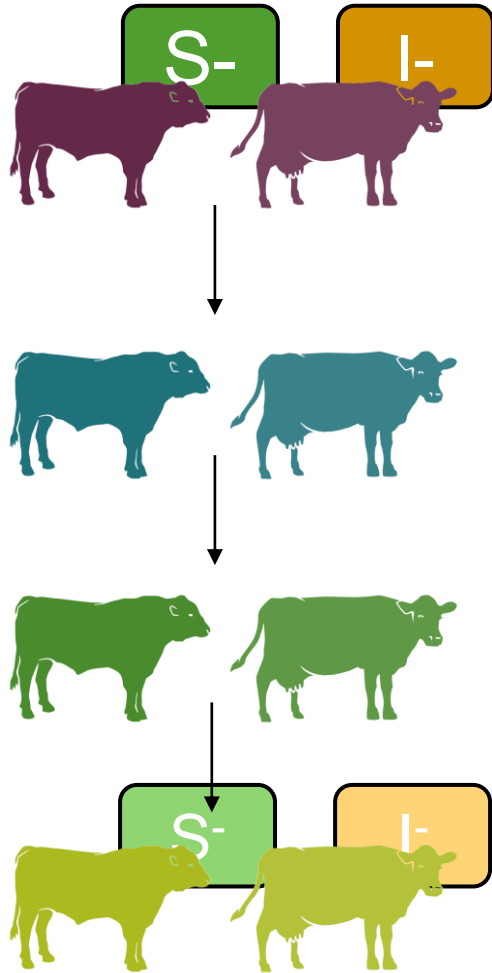


- **Individual-based models :**

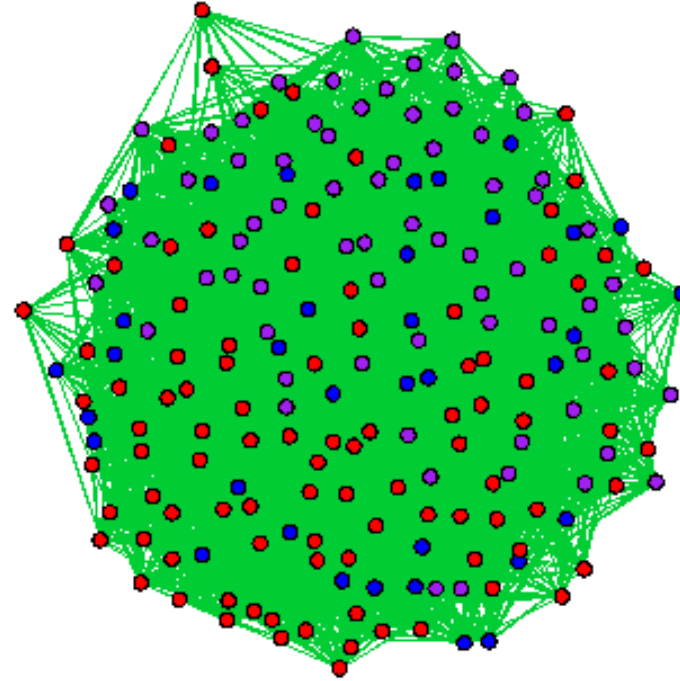
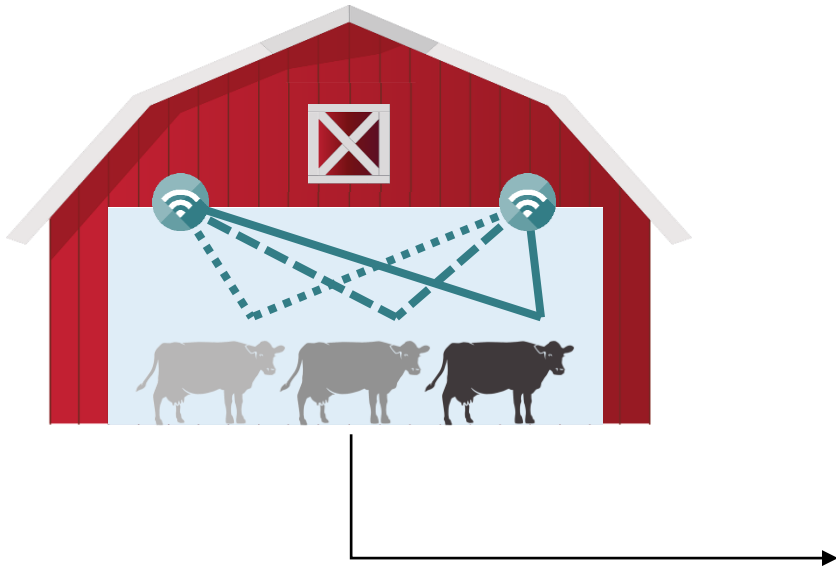
Represents each individual explicitly and tracks its state and characteristics over time.



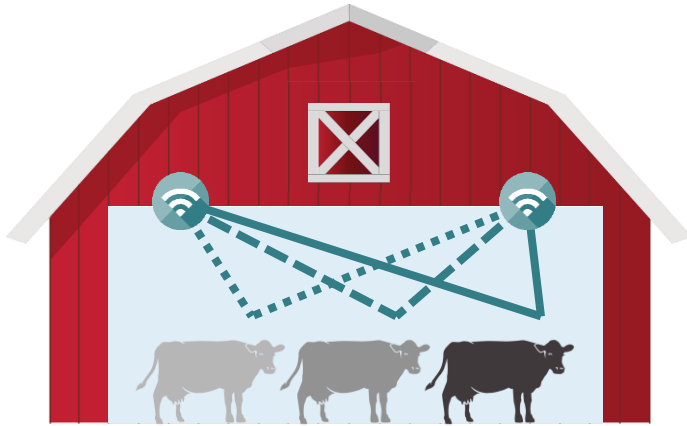
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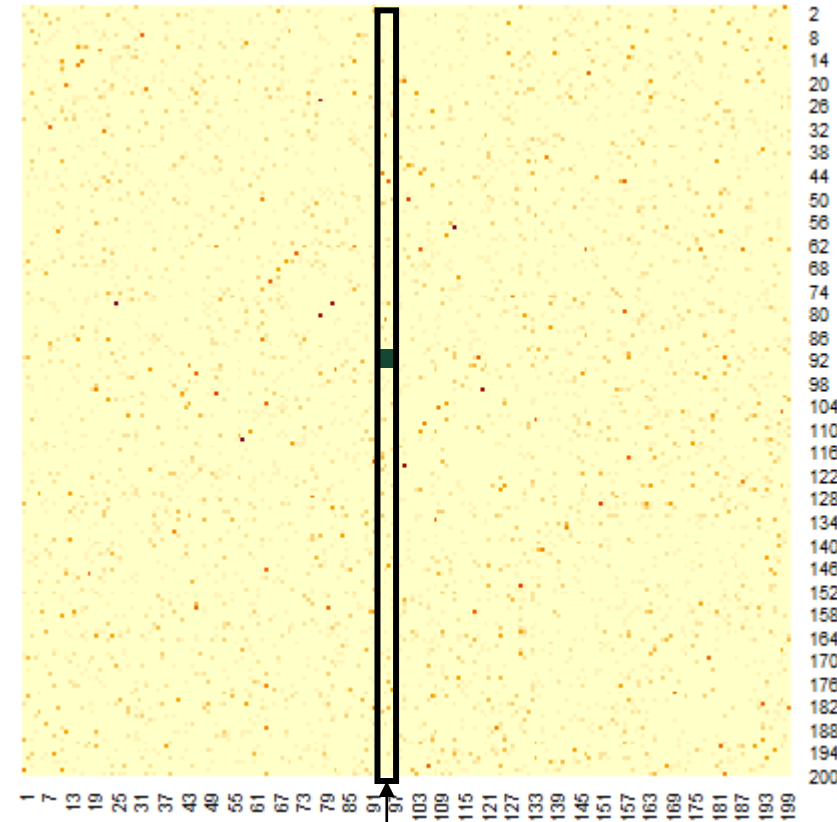


3. Incorporating breeding values into transmission models



$$p_{ij} = 1 - e^{-\beta \phi_{ij,t}}$$

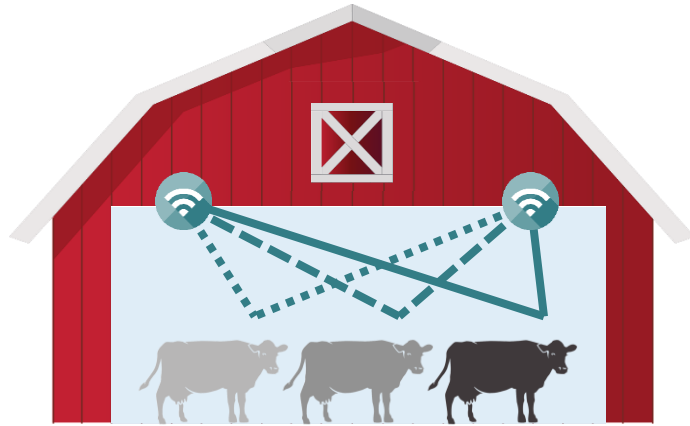
$$P(\text{infection})_{it} = 1 - \prod_{j=1}^k (1 - p_{ij,t})$$



3. Incorporating breeding values into transmission models

$$\phi_{i,t} * w_S \begin{cases} w_S > 1 \rightarrow S^+ \\ w_S < 1 \rightarrow S^- \end{cases}$$

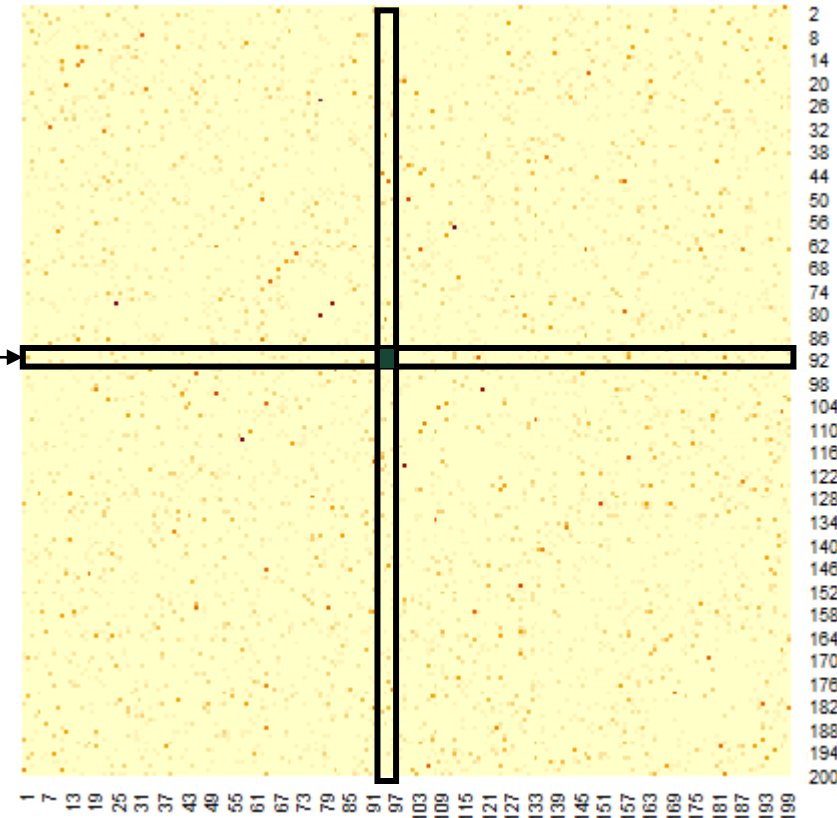
$$S_{i=91} = 0$$



$$I_{i=91} = 0$$

$$p_{ij} = 1 - e^{-\beta \phi_{ij,t}}$$

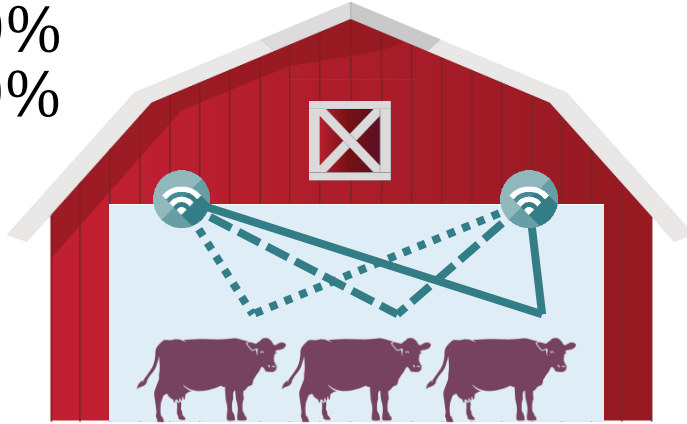
$$\phi_{j,t} * w_I \begin{cases} w_I > 1 \rightarrow I^+ \\ w_I < 1 \rightarrow I^- \end{cases}$$



3. Incorporating breeding values into transmission models

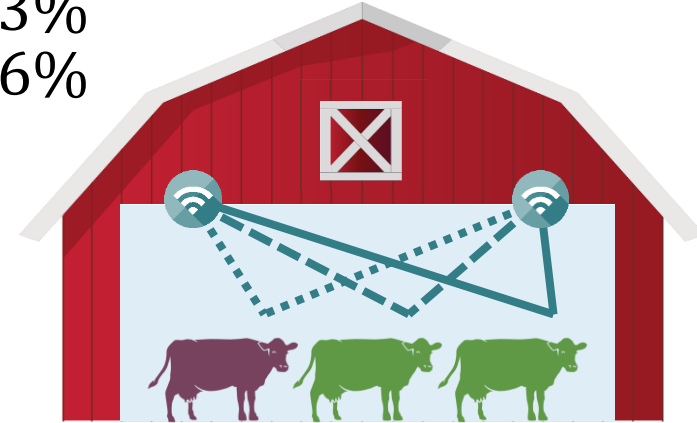
$$S^+I^+ = 100\%$$

$$S^-I^- = 0\%$$



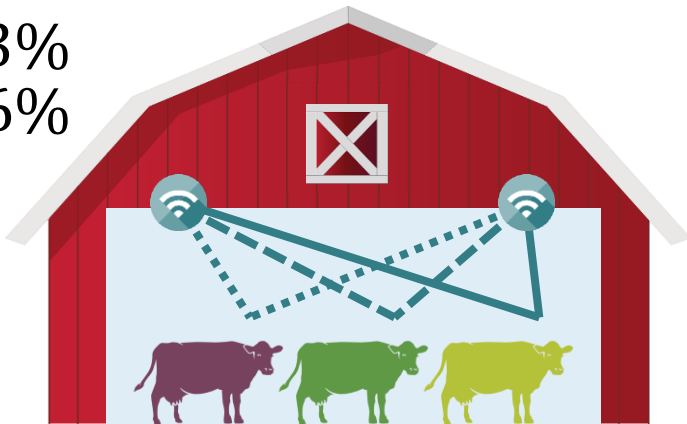
$$S^+I^+ = 33\%$$

$$S^-I^- = 66\%$$



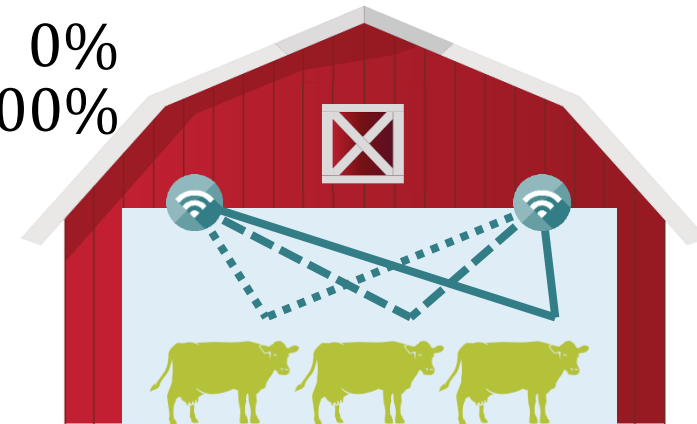
$$S^+I^+ = 33\%$$

$$S^-I^- = 66\%$$



$$S^+I^+ = 0\%$$

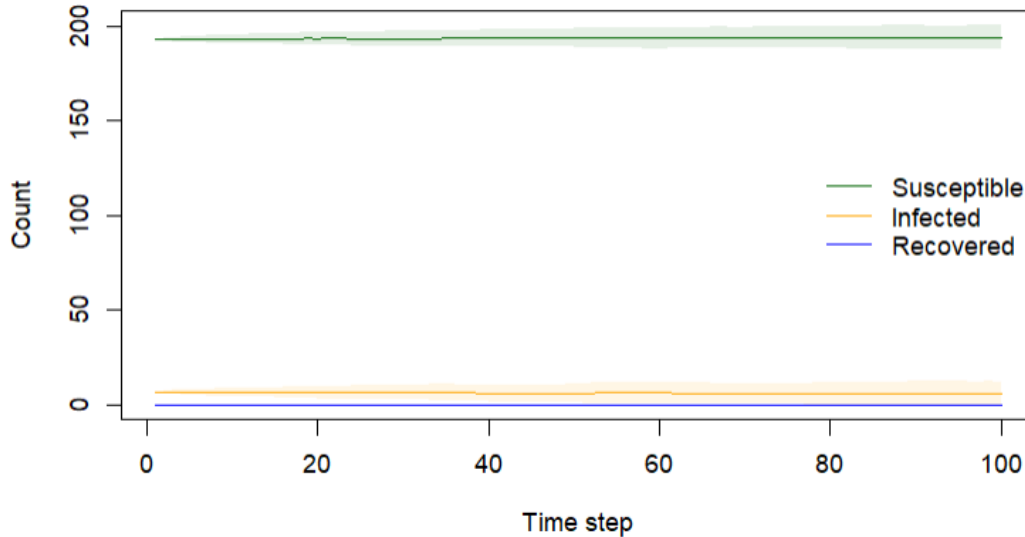
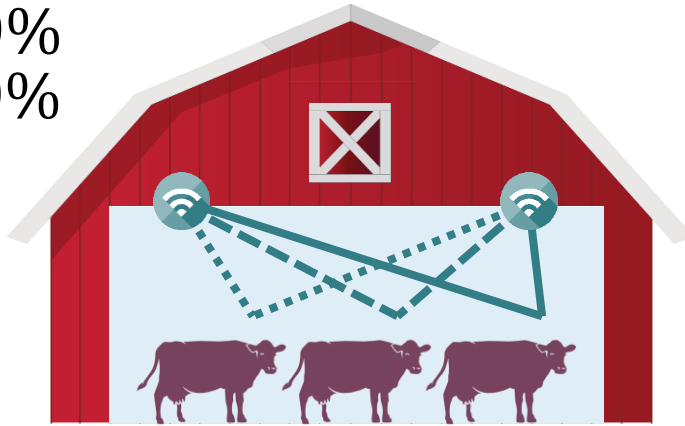
$$S^-I^- = 100\%$$



3. Incorporating breeding values into transmission models

$$S^+I^+ = 100\%$$

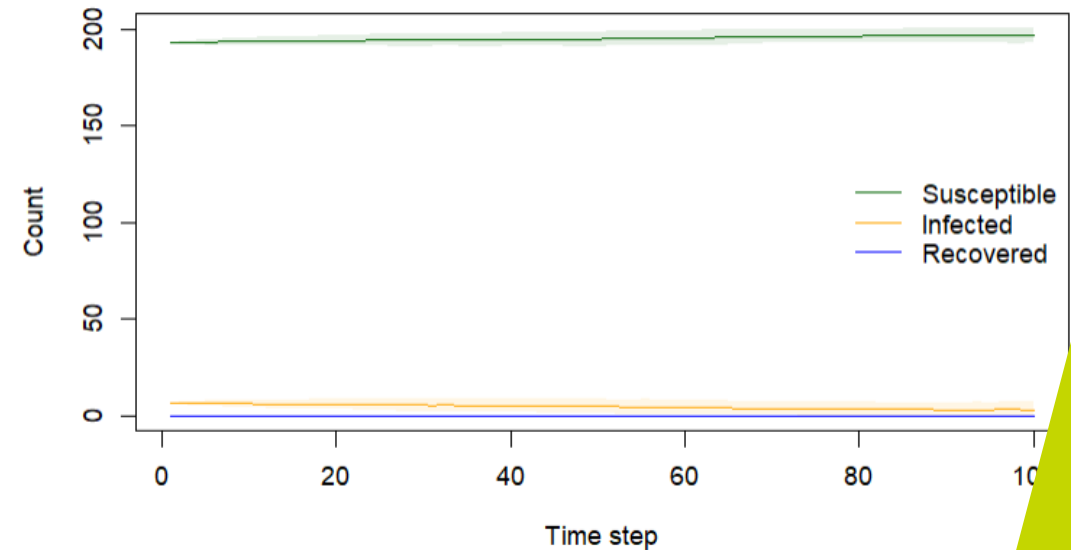
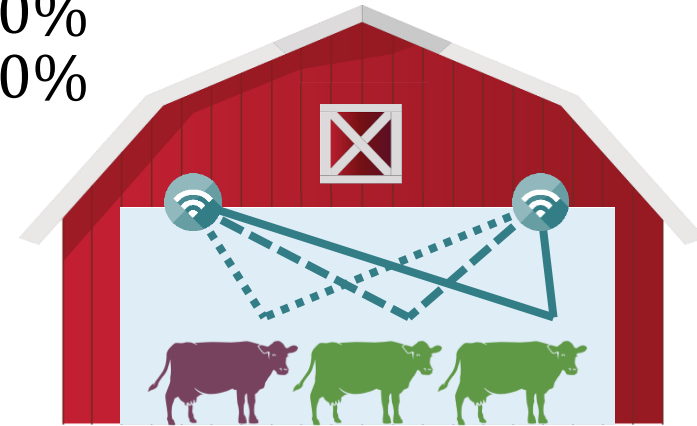
$$S^-I^- = 0\%$$



Prevalence = 3.14

$$S^+I^+ = 20\%$$

$$S^-I^- = 70\%$$



Prevalence = 2.40

Learning objectives

1. *Understand different frameworks for simulating disease transmission.*
2. *Implement compartmental and stochastic transmission models.*
3. *Parameterize transmission models using real data.*
4. *Incorporate spatial information into transmission models.*
5. *Incorporate breeding values into transmission models.*

Resources

- *Essential*

- Keeling, M.J., Rohani, P., 2011. Modeling infectious diseases in humans and animals, Modeling Infectious Diseases in Humans and Animals. [https://doi.org/10.1016/s1473-3099\(08\)70147-6](https://doi.org/10.1016/s1473-3099(08)70147-6)
- Vynnycky, E., White, R.G., 2010. An introduction to infectious disease modelling. Oxford University Press, New York. <https://doi.org/10.1093/eurpub/ckq163>

- *Further*

- Marina, H., Sanchez, J., Woudstra, S., Doeschl-Wilson, A., Nielsen, P. P., Nyman, A., & Rønnegård, L. (2026). Utilizing data from real-time location systems to find the most likely network of transmission given observed infections. *In preparation*. <https://github.com/Hector-Marina/SimNet>
- Pooley, C.M., Marion, G., Prentice, J. et al. 2025. SIRE 2.0: a novel method for estimating polygenic host effects underlying infectious disease transmission, and analytical expressions for prediction accuracies. *Genet Sel Evol.* <https://doi.org/10.1186/s12711-025-00956-4>
- de Freslon, I., Martínez-López, B., Belkhiria, J., Strappini, A., Monti, G., 2019. Use of social network analysis to improve the understanding of social behaviour in dairy cattle and its impact on disease transmission. *Appl. Anim. Behav. Sci.* <https://doi.org/10.1016/J.APPLANIM.2019.01.006>
- Egan, M.E., Pepin, K.M., Fischer, J.W., et. al. 2023. Social Network Analysis of White-Tailed Deer Scraping Behavior: Implications for Disease Transmission. *Ecosphere*. <https://doi.org/10.1002/ecs2.4434>
- Hirsch, B.T., Prange, S., Hauver, S.A., Gehrt, S.D. 2013. Raccoon Social Networks and the Potential for Disease Transmission. *PLoS ONE*. <https://doi.org/10.1371/journal.pone.0075830>
- Marina, H., Nielsen, P.P., Fikse, W.F., Rønnegård, L., 2024. Multiple factors shape social contacts in dairy cows. *Appl. Anim. Behav. Sci.* <https://doi.org/10.1016/J.APPLANIM.2024.106366>
- Doeschl-Wilson, A., Knap, P.W., Opriessnig, T., et al. (2021). Review: Livestock disease resilience: from individual to herd level. *Animal*. <https://doi.org/10.1016/j.animal.2021.100286>
- Center for Communicable Disease Dynamics. Harvard. T.H. Chan. Introduction to Infectious Disease Modeling. <https://hsph.harvard.edu/research/communicable-disease-ccdd/resources/id-modeling/>

hands
on





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